Project Proposal

CSE 1102

Structured Programming Laboratory

Team Information

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Proposal Details

1. Project Title: Integrated KUET Campus Management System

2. Short Description of the Project

The system will be a console-based menu-driven campus management system developed in C. When the program starts, the user will log in as a Student, Teacher, or Administrator.

A Student will be able to view **personal details** (name, department, roll, contact info), **hall information** (hall name, boarder number, residential allocation, meal management, upcoming feasts), **academic records** (registered courses with credits, course-wise attendance graphs, advisor, CGPA), and **library information** (books borrowed, issue/return dates).

A Teacher will be able to view and update their **teacher profile** (name, ID, designation, department, contact info, qualifications, research specialization, short bio), **search** for students, view their **academic records**, and **update marks** (class test and term examination) by processing **input files**.

An Administrator will manage the **overall system** records. The administrator will be able to **add, edit, and remove student and teacher records**, publish **notices**, and **oversee system data**.

The general workflow will be: Start Program → Select Role → Login → Role-Based Menu → Perform Operation → Save/Update Data → Logout

3. Lab Topic Integration

- **Control Structures:** Implementing login validation, menu-driven navigation, input validation, searching, and record processing.
- **Functions & Modular Design:** Creating separate functions for login, student management, teacher management, course management, result calculation, notice management, and file operations; improving workflow, debugging, and unit-testing.
- **Arrays:** Storing marks, attendance, course lists, and routine information.
- **Strings:** Managing student and teacher names, IDs, course codes, passwords, complaints, and notices.
- **Structures & typedef:** Defining structured records for students, teachers, courses, complaints, and notices.
- **Pointers & Pointer Arithmetic:** Passing structures to functions, modifying records through pointers, and traversing dynamically allocated data.
- **Dynamic Memory Allocation:** Dynamically storing a variable number of students, courses, notices, and other records.
- **File Handling:** Storing, reading, updating, and exporting student, teacher, course, complaint, and notice records using files.
- **Command-Line Arguments:** Passing startup options or configuration information when launching the program.
- **Multiple Files & Header Files:** Dividing the project into multiple .c and .h files to keep the system organized and modular.
- **Makefile:** Automating the compilation process when working with multiple files.

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